Mock exam 1: solutions and marking guide

About this mock exam. This mock exam was written by an AI model, from written instructions on how the exam is structured and what it should examine. It is based on the ordinary exam of 2025, adjusted for length and difficulty: that exam was a 12-hour take-home exam with every aid allowed, while this year's exam is three hours at the department, without aids.

Some of it is on lectures still to come. Part A is on lecture 9 (network effects in green technology adoption, week 46), essay C.1 on lecture 10 (the energy efficiency gap, week 47) and essay C.3 on lecture 13 (energy and geopolitics, week 50). Part B (lectures 2, 4 and 6) and essay C.2 (lecture 5) are on material you have already had.

The solutions, with the marking guide for every question, are in a separate PDF. Try the exam first.

How to read the solutions. An answer needs no references and no numbers to earn full marks, unless the question asks for them. The solutions below cite both where they help the reader; a student's answer does not have to.

Each question has a solution and a marking guide. The guide lists the points an answer must make, with the marks each carries, mechanisms first; a point marked *supporting* strengthens an answer but is not needed for full marks. It ends with what an answer at each level does.

Part A. Electric vehicles and charging stations 35 points

A.1 4 points

Explain the difference between a direct and an indirect network effect, and why the relation between EVs and charging stations is an indirect one. Why might it be a reason for policy to intervene?

Solution. A network effect is present when one agent's use of a good raises its value to other users. It is *direct* when other users' adoption enters a user's utility itself (a telephone network, or neighbours' solar panels in Bollinger and Gillingham 2012). It is *indirect* when the benefit runs through a complementary good: more users make it profitable to supply more of the complement, and the complement makes the good more valuable. EVs and chargers are of the second kind, the hardware-software pattern: more EVs raise the return to building a charger, and more chargers raise the value of owning an EV. A driver gains from other EV owners only because they bring chargers.

It can justify intervention because neither side internalises its effect on the other: a buyer ignores that their purchase makes the next charger more profitable, and a station owner ignores what their charger adds to other drivers' value of an EV, so adoption can be too low. Whether it is a market failure depends on whether those effects are priced, which the model in this part does not settle.

Marking guide.

- An indirect effect runs through a complementary good, a direct one through users' utility (1.5 points)
- EVs raise the return to chargers and chargers raise the value of EVs, a two-sided feedback (1.5 points)
- Neither side internalises its effect on the other, so adoption may be too low (1 point)

Full marks. Both definitions, the feedback in both directions, and the unpriced effect as the case for policy.

About half. The feedback described, without the distinction from a direct effect or without the policy case.

No marks. Calls it a direct effect, or describes only one direction.

A.2

5 points

Interpret β_1, β_2 and γ_1. Then show that the stock of EVs evolves according to

$$Q_{t+1} = f(Q_t) \equiv Q_t^{\beta_1\gamma_1}\, p_t^{\beta_2} + \delta\, Q_t. \qquad \text{(A4)}$$

Solution. β_1 is the elasticity of EV sales with respect to the number of stations: a one per cent larger network raises sales by β_1 per cent. β_2 is the own-price elasticity of sales, negative. γ_1 is the elasticity of the network with respect to the stock of EVs: how strongly station owners respond to a larger fleet. Both equations are log-linear, so the elasticities are constant.

Taking exponentials of (A2) gives $N_t = Q_t^{\gamma_1}$. Substituting into (A1) and taking exponentials gives $q_t = N_t^{\beta_1} p_t^{\beta_2} = Q_t^{\beta_1\gamma_1} p_t^{\beta_2}$, and substituting this into (A3) gives (A4).

Marking guide.

- β_1 and γ_1 as the two elasticities of the feedback, β_2 as the own-price elasticity (2 points)
- $N_t = Q_t^{\gamma_1}$ substituted into (A1), giving $q_t = Q_t^{\beta_1\gamma_1} p_t^{\beta_2}$ (2 points)
- Substituted into (A3) to give (A4) (1 point)

Full marks. All three parameters read as elasticities, and the two substitutions.

About half. The derivation without the interpretation, or the reverse.

No marks. Neither.

A.3

8 points

Let the EV price be constant, $p_t = p$. Derive the steady state $Q^* > 0$. Show that the slope of the transition function at the steady state is

$$f'(Q^*) = \beta_1\gamma_1(1-\delta) + \delta,$$

and use it to state the condition under which Q^* is stable. Draw the transition diagram for this case, and explain in words what the condition says about the feedback between cars and chargers. Does the model have a critical mass?

Solution. In a steady state $Q_{t+1} = Q_t = Q^*$. From (A4), $Q^*(1-\delta) = (Q^*)^{\beta_1\gamma_1}p^{\beta_2}$, and for $Q^* > 0$

$$Q^* = \left(\frac{p^{\beta_2}}{1-\delta}\right)^{1/(1-\beta_1\gamma_1)}.$$

$Q = 0$ is a steady state too.

The slope is $f'(Q) = \beta_1\gamma_1 Q^{\beta_1\gamma_1 - 1}p^{\beta_2} + \delta$. At the steady state, $(Q^*)^{\beta_1\gamma_1-1}p^{\beta_2} = 1-\delta$, so $f'(Q^*) = \beta_1\gamma_1(1-\delta) + \delta$. This lies between 0 and 1, so that Q^* is stable, if and only if $\beta_1\gamma_1 < 1$. Then f is also concave, steep near zero and flatter than the 45-degree line beyond Q^*: from any $Q_0 > 0$ the stock converges monotonically to Q^*, from below or from above.

The transition diagram has Q_{t+1} against Q_t: a concave curve that starts at the origin with an infinite slope and crosses the 45-degree line once, from above, at Q^*; the staircase from a small Q_0 climbs to Q^*.

$\beta_1\gamma_1$ is the strength of the feedback: a one per cent larger fleet raises the network by γ_1 per cent, which raises sales by $\beta_1\gamma_1$ per cent. If that is less than one, each round of the loop is weaker than the last, so the feedback amplifies a shock but dies out, as a geometric series converges.

There is no critical mass. Every positive starting stock converges to the same Q^*, and $Q = 0$ is unstable: the model has no threshold below which adoption collapses and above which it takes off.

Marking guide.

- The steady state Q^* derived, with $Q = 0$ noted as a second steady state (2 points)
- The slope at Q^* derived, and stability if and only if $\beta_1\gamma_1 < 1$ (2 points)
- A transition diagram with a concave curve crossing the 45-degree line once (1.5 points)
- The condition read as a feedback that dies out, each round weaker than the last (1.5 points)

- No critical mass, because every positive start converges and zero is unstable (1 point)

Full marks. Steady state, slope and stability condition, a correct diagram, the feedback read as dying out, and no critical mass with the reason.

About half. The steady state and the condition, with the diagram or the interpretation missing.

No marks. No steady state, or stability claimed without the condition.

A.4 7 points

Assume the condition from A.3 holds. Show that the long-run elasticity of EV sales with respect to the price, in the steady state, is

$$\frac{d\ln q^*}{d\ln p} = \frac{\beta_2}{1-\beta_1\gamma_1},$$

and compare it with the elasticity of sales in a single period, holding the stock of stations fixed. What does the comparison say about the effect of a permanent purchase subsidy? And what does a subsidy paid only for a few periods do to the path of Q_t and to the steady state?

Solution. In the steady state $q^* = (1-\delta)Q^*$, so, with A.3,

$$\ln q^* = \ln(1-\delta) + \frac{\beta_2 \ln p - \ln(1-\delta)}{1-\beta_1\gamma_1}, \qquad \frac{d\ln q^*}{d\ln p} = \frac{\beta_2}{1-\beta_1\gamma_1}.$$

Within a period, with the stock and so the network given, the elasticity is β_2, from (A1). With $0 < \beta_1\gamma_1 < 1$ the long-run elasticity is larger in absolute value by the factor $1/(1-\beta_1\gamma_1)$: a lower price sells more cars, which bring more stations, which sell more cars, and the series sums to that factor.

A permanent purchase subsidy therefore does more in the long run than its direct effect in the year it is paid. A subsidy paid for a few periods raises sales while it lasts, and since the larger stock brings more stations, sales stay above the path without it after it ends. But the steady state depends only on the permanent price, so the stock converges back to the same Q^*: a temporary subsidy brings adoption forward and does not change where it ends.

Marking guide.

- The long-run elasticity derived from $q^* = (1-\delta)Q^*$ and Q^* (2 points)
- The short-run elasticity β_2, and the long-run one larger by $1/(1-\beta_1\gamma_1)$ because of the feedback (2 points)
- A permanent subsidy does more in the long run than its direct effect (1 point)
- A temporary subsidy speeds adoption, keeps sales up after it ends, and leaves the steady state unchanged (2 points)

Full marks. The derivation, the amplification with its mechanism, and both subsidies.

About half. The elasticity without the comparison, or the temporary subsidy wrong (a permanent effect).

No marks. No derivation and no mechanism.

A.5 5 points

Now suppose the feedback is strong: $\beta_1\gamma_1 > 1$. Draw the transition diagram, and say which steady states are stable. Does the model now have a critical mass? What does your answer imply for a subsidy paid only for a few periods?

Solution. With $\beta_1\gamma_1 > 1$ the transition function is convex. Its slope at zero is $\delta < 1$, and at the positive steady state it is $\beta_1\gamma_1(1-\delta)+\delta > 1$ (A.3). So $Q = 0$ is stable and Q^* is unstable. Since $f(Q)/Q = Q^{\beta_1\gamma_1-1}p^{\beta_2} + \delta$ rises with Q and equals one at Q^*, the curve lies below the 45-degree line to the left of Q^* and above it to the right: a stock below Q^* shrinks towards zero, and a stock above it grows without bound.

Q^* is now a critical mass: the point beyond which adoption becomes self-reinforcing. A temporary subsidy then works very differently. One that lifts the stock above Q^* has a permanent effect, because adoption carries on after it ends; one that falls short is wasted, since the stock slides back. Timing and size matter, and a large, short push can beat a small, long one. Li et al.'s estimates put the feedback below one, which is the case of A.3.

Marking guide.

- Zero stable and Q^* unstable, from the slopes or the diagram (2 points)
- Below Q^* the stock falls to zero, above it adoption takes off, so Q^* is a critical mass (1.5 points)
- A temporary subsidy that lifts the stock past Q^* has a permanent effect; one that does not is wasted (1.5 points)

Full marks. Stability of both steady states, the critical mass with its reason, and the consequence for a temporary subsidy.

About half. The critical mass named, with the stability or the policy consequence missing.

No marks. Claims the dynamics are as in A.3.

A.6 6 points

Li et al. (2017) find that the US federal tax credit for EV purchases could have bought considerably more EVs had the same money been spent on subsidising charging stations. Using the model, explain what makes a station subsidy more cost-effective than a purchase subsidy. Would you expect the conclusion to carry over to Denmark today, where EVs make up most new car sales? Does the paper's result show that subsidising charging stations is socially optimal?

Solution. A purchase subsidy acts on sales directly, through β_2, and on the network only through the feedback. A station subsidy adds stations directly, and each station raises sales through the network elasticity β_1. The station subsidy is the more cost-effective when sales respond strongly to the network relative to how strongly the network responds to the fleet, and when sales respond weakly to the price: early buyers who would have bought anyway collect a purchase subsidy without changing their decision. In Li et al. the network effect on demand is the stronger of the two, and early EV buyers were not very price-sensitive.

In Denmark today the conditions differ. EVs make up most new car sales, ranges are longer and many owners charge at home, so an extra public charger probably adds less to the value of an EV than in the US of 2011-13, and the marginal buyer is a mainstream buyer who is more price-sensitive. Both point towards a smaller advantage for station subsidies; Li et al. themselves note that their ranking depends on these features.

The result is about cost-effectiveness: it takes the goal of more EVs as given and asks which subsidy reaches it more cheaply. It does not show that subsidising either side is optimal, which would require the network effect to be an unpriced externality and a comparison of the subsidy's cost with the value of what it achieves.

Marking guide.

- Station subsidies win when sales respond strongly to the network and weakly to the price, and when inframarginal buyers capture purchase subsidies (2.5 points)
- In Denmark today (mainstream buyers, longer range, home charging) the advantage is probably smaller (2 points)
- The paper is about cost-effectiveness for a given goal, not about optimality (1.5 points)

Full marks. The mechanism in the model's terms, a reasoned view on Denmark, and cost-effectiveness distinguished from optimality.

About half. Two of the three.

No marks. Restates the finding without the mechanism.

Part B. Battery storage in a market with wind and solar 35 points

B.1 6 points

For a given power rating and duration, write down the firm's problem of choosing charging, discharging and stored energy in every hour to maximise its operating profit over the year. Explain each constraint in a line.

Solution.

$$\max_{g^c_h,\, g^d_h,\, soc_h} \sum_h \Big[\lambda_h \,(g^d_h - g^c_h) - c^s \,(g^c_h + g^d_h)\Big]$$

subject to, for every hour h,

$$\begin{aligned} soc_h &= soc_{h-1} + \eta_c \, g^c_h - g^d_h/\eta_d, \\ 0 \le g^c_h &\le \bar{s}, \qquad 0 \le g^d_h \le \bar{s}, \\ 0 &\le soc_h \le \bar{\delta}\,\bar{s}, \\ soc_T &= soc_0. \end{aligned}$$

The objective is revenue from selling minus the cost of buying, minus wear. The law of motion adds what charging puts in, after the charging loss, and removes what discharging takes out, which exceeds what reaches the grid. The two power limits cap charging and discharging at the rating, and the energy limit caps what is held at the rating times the duration. The cyclic condition stops the battery from selling energy it never bought by emptying itself at the end of the year.

Marking guide.

- The objective, revenue minus charging cost minus wear (1.5 points)
- The law of motion with both efficiencies in the right places (2 points)
- Power limits on charging and discharging, and the energy limit (1.5 points)
- The cyclic condition, with its reason (1 point)

Full marks. All constraints right, each explained.

About half. The objective and the law of motion right, a limit or the cyclic condition missing.

No marks. No law of motion.

B.2 7 points

With θ_h the multiplier on the law of motion for stored energy, $\underline{\phi}^c_h, \overline{\phi}^c_h$ and $\underline{\phi}^d_h, \overline{\phi}^d_h$ those on the charging and discharging limits, and $\underline{\psi}_h, \overline{\psi}_h$ those on the energy limit, the first-order conditions are

$$\begin{aligned} g^c_h &: \; -c^s - \lambda_h + \eta_c\,\theta_h + \underline{\phi}^c_h - \overline{\phi}^c_h = 0, \\ g^d_h &: \; -c^s + \lambda_h - \theta_h/\eta_d + \underline{\phi}^d_h - \overline{\phi}^d_h = 0, \\ soc_h &: \; -\theta_h + \theta_{h+1} + \underline{\psi}_h - \overline{\psi}_h = 0. \end{aligned}$$

What does θ_h measure? Use the conditions to explain in which hours the battery charges, discharges and does neither, and why θ_h can change from one hour to the next only when a limit on stored energy binds.

Solution. θ_h is the value to the firm of one more MWh in the battery at the end of hour h: the shadow price of stored energy, a stock rather than a flow.

Read the charging condition at its bounds. The battery charges when $\lambda_h \leq \eta_c \theta_h - c^s$: flat out when the price is strictly below, where $\overline{\phi}_h^c > 0$ is the rent on the power rating, and not at all when the price is above. Discharging mirrors it: the battery discharges when $\lambda_h \geq \theta_h/\eta_d + c^s$. Between the two thresholds it holds. This is the arbitrage band of ESM section 4.1; its width is the round-trip loss plus twice the wear cost, so the battery never charges and discharges in the same hour.

The condition for soc_h says $\theta_{h+1} = \theta_h$ unless $\underline{\psi}_h$ or $\overline{\psi}_h$ is positive, which happens only when the battery is empty or full. When it is neither, a marginal MWh can be moved from one hour to the next in either direction, so its value cannot differ between them. When it is full, θ jumps up, $\theta_{h+1} = \theta_h + \overline{\psi}_h$: energy would be worth more later, but there is no room to hold more of it.

Marking guide.

- θ_h as the value of a MWh in store (1.5 points)
- Charge below $\eta_c \theta_h - c^s$, discharge above $\theta_h/\eta_d + c^s$, hold in between, read from the bounds (3 points)
- The band's width is the round-trip loss and twice the wear cost (1 point)
- θ_h constant unless the battery is empty or full, because the MWh can then be moved either way (1.5 points)

Full marks. The meaning of θ_h, the three regimes from the conditions, the band's width, and why θ_h moves only at a full or empty battery.

About half. The regimes stated without deriving them from the bounds, or θ_h unexplained.

No marks. No regimes.

B.3 6 points

The firm's battery is small, but suppose many batteries enter the zone. What do they do to the distribution of prices over the year, and what does that do to what the next battery can earn? Would a planner who chose the batteries' operation to minimise system cost operate them differently from price-taking firms?

Solution. Batteries buy in the cheapest hours and sell in the dearest. Many of them raise the low prices and lower the high ones: the price duration curve is compressed from both ends, towards the band in which no battery wants to trade (ESM section 4.2). The spreads a battery lives on shrink, so each additional battery earns less than the last. With free entry, batteries enter until the next one would earn just its fixed cost.

A planner who minimised system cost would operate them the same way. The firm's conditions are the planner's (ESM section 4.1), with the market price in place of the planner's dual λ_h: price-taking arbitrage moves energy from hours where it is cheap to hours where it is dear exactly as far as the planner would, so the competitive outcome is efficient. The exception is a battery large enough to move prices: it would trade less than the planner, to keep the spreads it earns on.

Marking guide.

- Entry compresses the price distribution from both ends (2 points)
- So each additional battery earns less, and entry stops where the next would just cover its cost (2 points)
- Price-taking operation is the planner's, the same conditions with the market price as the dual (2 points)
- Unless the batteries have market power, and then they trade too little (*supporting*)

Full marks. Compression, falling earnings and where entry stops, and efficiency of price-taking operation with its reason.

About half. Two of the three.

No marks. Claims batteries make prices more volatile.

B.4

6 points

Now let the firm also choose the power rating $\bar{s} \geq 0$ (multiplier $\underline{\beta}_s$) of a battery with a given duration $\bar{\delta}$, at an annual fixed cost F_s per MW. The first-order condition for $\bar{s}$ is

$$-F_s + \sum_h \left(\overline{\phi}^c_h + \overline{\phi}^d_h\right) + \bar{\delta} \sum_h \overline{\psi}_h + \underline{\beta}_s = 0.$$

Interpret the condition for a battery that is built. In which hours does a MW of battery earn its keep, and what decides how much it earns?

Solution. For a battery that is built, $\bar{s} > 0$ and $\underline{\beta}_s = 0$, so

$$\sum_h \left(\overline{\phi}^c_h + \overline{\phi}^d_h\right) + \bar{\delta} \sum_h \overline{\psi}_h = F_s :$$

another MW earns exactly its annual fixed cost, in rents on its limits. It earns on power in the hours it charges or discharges at the rating, when the price is outside the band and more power would move more energy from cheap hours to dear ones; and on energy, counted $\bar{\delta}$ times because a MW comes with $\bar{\delta}$ MWh, in the hours it is full, when more room would carry more cheap energy to dearer hours.

Every one of these rents is a price difference net of losses and wear. So what a MW earns is decided by how wide the price swings are, how often they come and how long they

last, not by any running cost: the battery is paid by volatility, and the condition is the battery's version of long-run zero profit (ESM Result 6).

Marking guide.

- With $\underline{\beta}_s = 0$, a built MW earns exactly its fixed cost in rents (2 points)
- Power rents at full charging or discharging, energy rents when full, counted $\bar{\delta}$ times (2 points)
- The rents are price spreads, so earnings depend on how wide, frequent and long the swings are (2 points)

Full marks. Zero profit for the built battery, both rents with their hours, and the rents read as the price swings.

About half. Zero profit stated, the rents not tied to hours or to the price swings.

No marks. Reads F_s as a running cost, or the condition as an inequality for a built battery.

B.5 5 points

In the greenfield model of ESM section 7, the planner can build batteries of 2, 4 or 8 hours' duration, and the reference case builds only 8-hour batteries. Figure 1 shows the median DK2 price in each hour of the day, month by month, in that case. Use the figure and the condition from B.4 to explain why the longest battery is the one built. Which duration would you expect to be built if the price instead spiked for a single hour each evening and were flat otherwise?

Solution. From May to August the DK2 price is close to zero through the middle of the day, when solar produces, and back at its usual level through the evening and the night; in April and September the midday dip is shallower, and through the winter the price is much the same at every hour. The cheap stretch and the expensive one each last many hours, so a battery can charge for many hours and discharge for many. What a MW earns then grows roughly with its hours of storage, through the energy rent $\bar{\delta}\sum_h \overline{\psi}_h$ of B.4. Its cost does not grow in proportion: the inverter costs the same per MW whatever the duration, and only the cells scale with it, so an hour of storage is cheapest in the longest battery. At the reference prices the 8-hour battery earns its fixed cost and the shorter ones would fall short of theirs (in the model a 2-hour battery would earn about seven tenths of its cost).

If the price instead spiked for one hour each evening and were flat otherwise, the extra hours of storage would earn nothing: there would be one expensive hour to sell into, and energy would be as cheap in any other hour. The shortest battery, which buys the power for the spike without paying for cells it cannot use, would be the one built, if any.

Marking guide.

- The swings in the figure are long, cheap through the middle of the day from May to August and back at the usual level through the evening and night (1.5 points)
- Earnings grow with duration while only part of the cost does, the inverter costing the same per MW (2 points)
- With a one-hour spike, extra hours earn nothing and the shortest battery is built (1.5 points)

Full marks. The long swings read from the figure, the cost structure, and the short battery for the short spike.

About half. The figure read, with the cost side or the extension missing.

No marks. A wrong reason: lower losses, or a lower cost per MW, of the long battery.

B.6

5 points

Suppose the grid company charges the battery a volumetric network tariff τ per MWh it draws from the grid, as it does any consumer. How does the tariff change the battery's operation and its investment decision? Argue whether, and when, such a tariff is justified.

Solution. A volumetric tariff on charging raises the cost of every MWh bought by τ: the battery now charges only when $\lambda_h + \tau \leq \eta_c \theta_h - c^s$. The arbitrage band widens, the battery trades in fewer hours and over smaller spreads, and its rents fall. Fewer batteries clear the investment condition of B.4, so less storage is built and less of the price volatility is smoothed. If the battery also pays a tariff on what it feeds in, it is charged twice for the same energy.

Whether this is justified depends on what the tariff is for. A network tariff should reflect the cost a user imposes on the network (cost-reflectivity, as in the regulation of network monopolies). A battery that charges in hours of surplus and discharges in hours of scarcity can relieve the grid rather than load it, and a flat volumetric tariff then charges it for a cost it does not cause and distorts its operation. A tariff that varied with network congestion, or a capacity-based charge, would charge the battery for the network cost it actually causes. A flat tariff can be defended only as a simple way to recover the network's fixed costs, which is an argument from simplicity and fairness rather than efficiency.

Marking guide.

- The tariff widens the band, so the battery trades less and earns less (1.5 points)
- Less investment, so less smoothing of prices (1 point)
- A tariff should reflect the network cost the battery causes, which may be negative when it charges in surplus hours (1.5 points)

- A time-varying or capacity-based charge as the alternative, or the fixed-cost recovery argument for a flat one (1 point)

Full marks. Operation and investment both affected, and a reasoned verdict on cost-reflectivity.

About half. The effect on operation and investment without a verdict, or the reverse.

No marks. Says the tariff has no effect because the battery sells the energy back.

Part C. Essays 30 points

C.1 10 points

> Engineering studies often find that energy efficiency investments would save money and cut emissions at the same time. Explain why evaluations made after a programme has run tend to find smaller gains than the projections made before it, and what this implies for subsidising energy efficiency as climate policy.

Solution. Ex ante projections come mostly from engineering models: the energy an investment would save under test conditions, valued at energy prices and set against its purchase cost. They leave out costs that households face but the model does not see (installation, disruption, a worse product in some dimension), tend to overstate savings, and assume an average user, while actual users differ in how much they use the good. People also use an efficient durable more, the rebound effect. Ex post evaluations measure what actually happened against a credible counterfactual; in the randomised evaluation of the US weatherisation programme by Fowlie, Greenstone and Wolfram (2018), realised savings fell well short of the projections and below the investment cost.

So the gap between private behaviour and cost-minimising investment may be much smaller than engineering curves suggest, and part of the "negative-cost abatement" is not there. A subsidy for efficiency then pays many people for what they would have done anyway and buys little abatement per krone. As climate policy it is a poor substitute for pricing the emission: a carbon price targets the externality and leaves people to choose how to save. A subsidy is justified only for a market failure in the investment decision itself, such as the landlord-tenant problem, and should be targeted at it (the targeting principle).

Marking guide.

- Ex ante engineering estimates omit unobserved costs, overstate savings and ignore heterogeneity (3 points)
- Rebound, or evidence that realised savings fall short of projections (2 points)

- The gap is smaller than claimed, so subsidies buy little abatement and pay infra-marginal adopters (2.5 points)
- Price the externality; subsidise only an identified investment failure, targeted at it (2.5 points)

Full marks. Why ex post falls short, with two or more reasons, and the policy implication through the targeting principle.

About half. Reasons without the policy implication, or the reverse.

No marks. Asserts a large free lunch.

C.2 10 points

> Do heat pumps help integrate wind power? Discuss what a growing fleet of heat pumps does to the cost of the electricity they use and to the market value of wind power, and what making the heat pumps flexible changes.

Solution. Heat pumps use most electricity in cold, dark hours, when demand is already high and so are prices. The average price they pay, weighted by their consumption, is their load cost, the mirror of wind's market value. As the heat pump fleet grows, its peak needs more peaking capacity, paid for through scarcity prices in exactly the hours heat pumps consume, so the load cost rises with their share (Ruhnau, Hirth and Praktiknjo 2020).

For wind the answer depends on what is held fixed. With wind capacity given, extra demand in windy winter hours raises wind's market value. But if wind grows with the new demand, to keep its share, the extra wind depresses the prices it earns again, and the gain largely disappears: the match between heat demand and wind is seasonal, not hourly. Heat pumps help wind much less than wind helps heat pumps.

Making heat pumps flexible, with thermal storage or system-friendly operation, shifts consumption out of the scarcest hours, which lowers the load cost. It does little for wind, because the new flexibility competes with the flexibility already in the system, such as trade and storage: flexibility crowds out flexibility.

Marking guide.

- The load cost as the consumption-weighted price, rising with the heat pump share because their peak needs scarcity-priced capacity (3.5 points)
- Heat pumps raise wind's value at fixed capacity but not at a fixed share, because the match is seasonal rather than hourly (3.5 points)
- Flexibility lowers the load cost but does little for wind, as it crowds out other flexibility (3 points)

Full marks. All three, each with its mechanism.

About half. Two of the three, or all three asserted without mechanisms.

No marks. Says heat pumps are flexible demand that solves wind's integration.

C.3 10 points

The G7 capped the price at which Russian oil may be transported with Western shipping and insurance. Explain why a lower cap need not reduce Russia's oil profits once Russia can build a shadow fleet, and what this implies for the design of sanctions.

Solution. Under the cap, Russia sells through two channels: with Western services at the capped price, or through its own shadow fleet at the world price less a discount. A lower cap makes the capped channel less profitable, so Russia moves exports to the shadow fleet and invests to expand it. While the fleet is too small to carry all exports, less oil reaches the world market and the world price rises, which raises what Russia earns on everything it ships through the fleet. That gain can offset the loss on the capped sales, so a lower cap, or even a complete service ban, can leave the present value of Russia's profits about the same or higher (Cardoso, Salant and Daubanes 2025), while oil consumers pay a higher world price.

For sanctions design, the evasion response has to be part of the analysis: a tighter instrument is not necessarily a more harmful one. A cap set too low invites investment in evasion and raises world prices. Measures aimed at the evasion channel itself, sanctions on the shadow fleet, raise its cost and work in the opposite direction. Timing matters too: the longer a cap is announced before it is enforced, the more time Russia has to build its fleet while selling at full prices.

Marking guide.

- A lower cap shifts exports to the shadow fleet and induces investment in it (2.5 points)
- While the fleet is constrained, world supply falls and the world price rises, raising fleet revenue and offsetting the capped loss (3.5 points)
- Design must account for evasion; a tighter cap is not necessarily more harmful, and it hurts consumers (2 points)
- Target the evasion channel (the fleet) or enforce early (2 points)

Full marks. The two channels, the price effect that offsets the capped loss, and design implications that follow from evasion.

About half. The shadow fleet as evasion, without the world-price mechanism.

No marks. Claims a lower cap always cuts Russia's profits.